

Lecture 2

Theorem: Let $a, b \in \mathbb{N}$. Then,

$$\gcd(a, b) \operatorname{lcm}[a, b] = ab.$$

Eg: Let $a = 12$ & $b = 9$.

$$\gcd(12, 9) = 3$$

$$\operatorname{lcm}[12, 9] = 36.$$

$$\gcd(12, 9) \operatorname{lcm}[12, 9] = 108 = 12 \times 9.$$

Proof: Let $d = \gcd(a, b)$ & $l = \operatorname{lcm}[a, b]$.

$$d \mid a \quad \text{and} \quad d \mid b.$$

$$\Rightarrow \frac{a}{d}, \frac{b}{d} \in \mathbb{N}.$$

$$\Rightarrow \frac{ab}{d} \in \mathbb{N}.$$

$$\text{claim} \quad \frac{ab}{d} = \operatorname{lcm}[a, b].$$

$$a \mid a \cdot \frac{b}{d} \quad \text{and} \quad b \mid \frac{a}{d} \cdot b.$$

Let $m \in \mathbb{N}$ be such that $a \mid m$ and $b \mid m$.

$$\text{we claim} \quad \frac{ab}{d} \mid m.$$

Since $d = \gcd(a, b)$, $\exists x, y \in \mathbb{Z}$ such that

$$d = ax + by.$$

$$\Rightarrow 1 = \frac{a}{d}x + \frac{b}{d}y \quad \Rightarrow \quad m = m \frac{a}{d}x + m \frac{b}{d}y.$$

$$\text{As } a|m \Rightarrow a \frac{b}{d} | m \frac{b}{d}.$$

$$\& b|m \Rightarrow \frac{a}{d} \cdot b | m \cdot \frac{a}{d}.$$

$$\Rightarrow \frac{ab}{d} | m \frac{a}{d} u + m \frac{b}{d} v = m.$$

$$\Rightarrow \frac{ab}{d} | m.$$

$$\Rightarrow \text{lcm}[a, b] = \frac{ab}{d}.$$

Euler phi function: Define $\phi: \mathbb{N} \rightarrow \mathbb{N}$ by

$$\phi(n) = \left| \left\{ a \in \mathbb{N} \mid 1 \leq a \leq n, \gcd(a, n) = 1 \right\} \right|$$

$$\forall n \in \mathbb{N}.$$

Here $|A|$ denotes the cardinality of the set A .

For each $n \in \mathbb{N}$, $\phi(n)$ denotes the number of positive integers less than n and coprime to n .

Ex.

$$\phi(1) = 1.$$

$$\phi(2) = 1$$

$$\phi(3) = 2$$

$$\phi(4) = 2$$

and so on.

1) If p is a prime number, then $\phi(p) = p-1$.

The converse is also true.

i.e. if $\phi(n) = n-1$

$\Rightarrow 1, 2, \dots, n-1$ are coprime to n .

$\Rightarrow n$ is a prime number.

2) If p is a prime number & $m \in \mathbb{N}$, then

$$\phi(p^m) = p^m - p^{m-1}$$

Proof: $\phi(p^m) = \# \text{numbers coprime to } p^m$.

$$\Rightarrow \phi(p^m) = p^m - |A|$$

where A is the set of all numbers x less equal to p^m , such that $\gcd(x, p^m) \neq 1$.

Since p is prime $\gcd(x, p^m) \neq 1$.

$$\Rightarrow p \mid x.$$

$$\text{So } A = \{p, 2p, 3p, \dots, p^{m-1} \cdot p = p^m\}$$

$$\therefore |A| = p^{m-1}$$

$$\Rightarrow \phi(p^m) = p^m - p^{m-1}$$

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3) If a and b are coprime integers, then

$$\phi(ab) = \phi(a) \phi(b).$$

Corollary: If $a_1, a_2, \dots, a_r$ are pairwise coprime ^{positive} integers, then $\phi(a_1 a_2 \dots a_r) = \phi(a_1) \phi(a_2) \dots \phi(a_r).$

Fundamental theorem of arithmetic: Every positive integer $n > 1$ can be uniquely expressed as a product of primes (not necessarily distinct) except for the order in which the primes appear.

$$n = q_1 q_2 \dots q_r \quad (q_i \text{ is prime for all } i)$$

$$= p_1^{k_1} \dots p_r^{k_r} \quad (p_1, p_2, \dots, p_r \text{ are distinct primes})$$

$$\text{then } \phi(n) = \phi(p_1^{k_1}) \phi(p_2^{k_2}) \dots \phi(p_r^{k_r})$$

$$= \prod_{i=1}^r \phi(p_i^{k_i})$$

$$= \prod_{i=1}^r (p_i^{k_i} - p_i^{k_i-1})$$

$$= \prod_{i=1}^r p_i^{k_i} \left(1 - \frac{1}{p_i}\right)$$

$$= n \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right)$$

Eg. ~~at~~ $n = 240 = 2^4 \times 3 \times 5$

$$\phi(240) = 240 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right)$$

$$= 240 \times \frac{1}{2} \times \frac{2}{3} \times \frac{4}{5}$$

$$= 64.$$

Congruence: Let n be a natural number. An integer a is said to be congruent to an integer b modulo n , if $n \mid a-b$.

It is written as $a \equiv b \pmod{n}$

Eg: $12 \equiv 8 \pmod{4}$

$$13 \not\equiv 2 \pmod{5}$$

Theorem: Let $a, b, c, d, n \in \mathbb{N}$. Then the following are true.

- (1) $a \equiv a \pmod{n}$
- (2) $a \equiv b \pmod{n} \Rightarrow b \equiv a \pmod{n}$
- (3) $a \equiv b \pmod{n}, b \equiv c \pmod{n}$ then $a \equiv c \pmod{n}$
- (4) $a \equiv b \pmod{n}, c \equiv d \pmod{n}$ then $a+c \equiv b+d \pmod{n}$
 $ac \equiv bd \pmod{n}$
- (5) $a \equiv b \pmod{n} \Rightarrow a^k \equiv b^k \pmod{n}$

Proof: (4) $a \equiv b \pmod{n}$ $c \equiv d \pmod{n}$

$$\Rightarrow n \mid a-b \quad \& \quad n \mid c-d$$

$$\Rightarrow n \mid (a-b) + (c-d) \Rightarrow n \mid (a+c) - (b+d)$$

$$\Rightarrow a+c \equiv b+d \pmod{n}$$

$$n \mid (a-b) \Rightarrow n \mid (a-b)c$$

$$n \mid (c-d) \Rightarrow n \mid (c-d)b$$

$$\Rightarrow n \mid (a-b)c + (c-d)b$$

$$\Rightarrow n \mid ac - bd$$

$$\Rightarrow ac \equiv bd \pmod{n}$$

(5) Proof by induction on k .
for $k=1$ it holds.

Let $a^{k-1} \equiv b^{k-1} \pmod{n}$

then by (4)

$$a \cdot a^{k-1} \equiv b \cdot b^{k-1} \pmod{n}$$

$$\Rightarrow a^k \equiv b^k \pmod{n}$$

Theorem: Two integers a and b are Congruent modulo n if and only if both a and b leave the same remainder when divided by n .

Proof: By division algorithm $\exists q_1, q_2, r_1, r_2$ such that

$$a = nq_1 + r_1 \quad \text{where } 0 \leq r_1 < n$$

$$b = nq_2 + r_2 \quad \text{where } 0 \leq r_2 < n$$

$$\text{So } a - b = n(q_1 - q_2) + (r_1 - r_2)$$

$$\Rightarrow n(q_1 - q_2) - (a - b) = r_2 - r_1$$

$$n \mid n(q_1 - q_2) \quad \& \quad n \mid a - b$$

$$\Rightarrow n \mid r_2 - r_1$$

$$\text{Since } r_1, r_2 < n \quad \text{we have } r_1 = r_2$$

Conversely if $r_1 = r_2$

$$\text{then } a - b = n(q_1 - q_2)$$

$$\Rightarrow n \mid a - b$$

$$\Rightarrow a \equiv b \pmod{n}$$

Relation on a Set: Let S be a non-empty set. A binary relation R on S is defined as a subset of $S \times S$. i.e. $R \subset S \times S$.

if $(a, b) \in R$ then we say aRb .

The relation R is said to be

- 1) Reflexive if $(a, a) \in R$. $\forall a \in S$
- 2) Symmetric if $(a, b) \in R \Rightarrow (b, a) \in R$. $\forall a, b \in S$
- 3) Antisymmetric if $(a, b) \in R$ & $(b, a) \in R \Rightarrow a = b$
 $\forall a, b \in S$
- 4) Transitive if $(a, b) \in R$, $(b, c) \in R \Rightarrow (a, c) \in R$.
 $\forall a, b, c \in S$.

A relation is said to be an equivalence relation if it is reflexive, symmetric and transitive.

A relation on S is said to be a partial order on S if it is reflexive, antisymmetric and transitive.

~~Equivalence~~

Relations are often denoted by ' $\sim$ '

Partial order are often denoted by ' $\leq$ '.

Equivalence class of an element $a = \{b \in S \mid aRb\}$.

Eg: Let $\mathbb{Z}$ be the set of ~~natural~~ numbers. Define a relation ' $\sim$ ' on $\mathbb{Z}$ by

$$a \sim b \text{ iff } a \equiv b \pmod{n} \quad (n \text{ is a fixed natural number})$$

Claim: ' $\sim$ ' is an equivalence relation on $\mathbb{Z}$.

- ① $a \sim a$ as $n | a - a = 0$.
- ② $a \sim b \Rightarrow b \sim a$.
- ③ $a \sim b \wedge b \sim c \Rightarrow a \equiv b \pmod{n} \wedge b \equiv c \pmod{n} \Rightarrow a \equiv c \pmod{n}$.

$\Rightarrow$ ' $\sim$ ' is an equivalence relation.

Let $a \in \mathbb{Z}$.

The equivalence class of a (denoted by $\bar{a}$)

$$= \{ n' \in \mathbb{Z} \mid n' \equiv a \pmod{n} \}$$

This is called the congruence class or residue class of a .

$$\bar{a} = \{ a + kn \mid k \in \mathbb{Z} \}$$

$$= \{ a, a \pm n, a \pm 2n, a \pm 3n, \dots \}$$

There are exactly n distinct equivalence classes mod n , namely

$$\bar{0}, \bar{1}, \dots, \overline{n-1}$$

the set of equivalence classes is denoted by

$$\mathbb{Z}/n\mathbb{Z} \text{ or } \mathbb{Z}_n.$$

One can define a notion of addition and multiplication on the elements of $\mathbb{Z}_n$ by

$$\bar{a} + \bar{b} = \overline{a+b} \quad \& \quad \bar{a} \cdot \bar{b} = \overline{ab}.$$

Theorem: The operation of addition and multiplication on $\mathbb{Z}_n$ is well-defined.

Proof: Let $\bar{a}_1 = \bar{a}_2$ and $\bar{b}_1 = \bar{b}_2$

~~Then~~ To prove: $\overline{a_1+b_1} = \overline{a_2+b_2}$

Since $\bar{a}_1 = \bar{a}_2$, $n \mid a_1 - a_2$

Similarly $n \mid b_1 - b_2$

$$\Rightarrow n \mid (a_1 - a_2) + (b_1 - b_2) = (a_1 + b_1) - (a_2 + b_2)$$

$$\Rightarrow a_1 + b_1 \equiv a_2 + b_2 \pmod{n}$$

Similarly $n \mid (a_1 - a_2) b_1$

$$n \mid (b_1 - b_2) a_2$$

$$\Rightarrow n \mid a_1 b_1 - a_2 b_2$$

$$\Rightarrow \overline{a_1 b_1} = \overline{a_2 b_2}$$

So well defined.

